Project: Go Home (2026)

"Get Home Alive."

BRANZINO: Kinematic Repurposing of Protective Equipment for Junctional Hemorrhage Control

1. Strategic Context and System Overview

In contemporary Tactical Combat Casualty Care (TCCC) environments, massive junctional hemorrhage (inguinal and axillary) remains untreatable via standard extremity tourniquets, while manual compression severely degrades survivability during active engagement.

This proposal details the kinematic repurposing of standard-issue infantry protective gear (Knee/Elbow Pads) into high-pressure hemostatic chamber walls. By integrating ultra-low-cost commodity materials (High-Tensile Agricultural PO/PE Film) with self-regulating physical valves, the system realizes a 'One-Shot & Run' doctrine. This approach minimizes cognitive load and overcomes battlefield entropy through automated pressure regulation and full-volume discharge.

2. Hardware Kinematic Specifications (Core Mechanisms)

The system operates without electronic sensors or actuators, relying entirely on mechanical geometry and material properties through a four-stage composite mechanism.

2.1. Constraint-Release Auto-Deploy Clip

- **Operating Principle:** In the standard configuration (impact-absorption mode), the expansion energy of integrated leaf springs (high-elasticity polymer) is physically constrained by the internal padding, preventing accidental activation.
- **Kinematic Transformation:** Upon the rapid manual removal of the shock-absorbing pad by a medic, the constrained tension is released, triggering the Auto-Deployment of the clips.
- **Function:** The deployed clips transform into 'Wide Latches,' which distribute mechanical load across a broad surface area to prevent Stress Concentration during coupling with the screening film.

2.2. Agricultural High-Tensile Film-Based Pressure Screen

- **Material Selection:** To circumvent the high cost and supply chain instability of specialized medical films, the system utilizes Agricultural Greenhouse Film (PO/PE family), characterized by extreme tensile strength and elongation capacity.
- **Integrated Sealing Mechanism:** The film features internal heat-sealed layers of easy-off gauze. The film acts as an impenetrable structural barrier, while pressure regulation is centralized through a single-path control valve located on the dorsal dome of the shield to eliminate design variables.

2.3. Cam-Over Sliding Triple Sequential V-Valve

The primary pressure regulation mechanism utilizes a springless Cam-Over Sliding Structure governed by plastic groove inclines and precise mechanical clearances.

- **V-Shape Geometric Design:** The triangular valve geometry creates a Wedge Effect during high-pressure injection. Even if foam residue adheres to the valve, the pressure forces an opening from the apex, ensuring Self-Cleaning and anti-stiction.
- **Differential Clearance Scaling:** Three valves are calibrated with varying sliding resistance and clearances (0.2–0.3 mm scale):
  - *Valve 1 (Low-Pressure):* Immediate sliding activation for initial gas venting and minor overpressure (0.3 mm clearance).
  - *Valve 2 (Standard Arterial):* Activates when pressure exceeds the threshold required for arterial hemorrhage control (0.2–0.3 mm clearance).
  - *Valve 3 (Emergency/Safety):* Final venting stage to prevent excessive pressure that would lead to secondary tissue necrosis.
- **Hard Limit Catch (Stopper):** A rigid mechanical stopper at the end of the sliding rail prevents a Blow-out of the valve assembly. This ensures that even at maximum vent capacity, the system physically preserves the Minimum Tamponade Pressure required for deep-seated hemostasis.

2.4. Closed-loop Docking Port & High-Pressure Purge

- **Docking Port:** An airtight interface located at the geometric center of the shield allows for the seamless coupling of the foam canister nozzle.
- **Self-Purging Effect:** If the V-valve clearances are obstructed by battlefield debris (e.g., mud), the hydraulic force of the initial full-can discharge strikes the internal chamber, forcibly ejecting contaminants through the valve orifices.

3. Tactical Execution Doctrine: One-Shot & Run

The hardware design aims to reduce the soldier's cognitive load to 0% during Care Under Fire (CUF).

1. **Pre-Purge:** The initial 0.5-second discharge of the primary spray clears debris from the clip grooves while applying a high-volume dose of antiseptic and anesthetic agents to the wound.
2. **Chamber Construction:** Following gauze application, the deployment clips are coupled with the film screen and secured to the anatomy via high-tensile ratchet adhesives.
3. **Full-Can Discharge:** The foam nozzle is docked, and the entire contents of the canister (100% volume) are discharged without the need for manual pressure titration.
4. **Automated Regulation & Egress:** The sequential valves mechanically detect and vent excess pressure. Once the medic identifies visual feedback (foam venting from the valves), they immediately extract the casualty (Shoot & Run).

4. Definitive Care and Removal Protocol

- **Torsional Shear Release:** Upon arrival at a Role 2/3 surgical facility, the surgeon applies a bilateral twisting motion to the external shield.
- **En Bloc Separation:** This induces Shear Force across the multi-layered gauze interface, allowing the entire foam plug to be separated En Bloc without inducing secondary tissue tearing. This facilitates immediate progression to definitive vascular repair and surgical intervention.

BRANZINO: Advanced Hemostatic and Tissue Management System

1. Mechanism of Gelation: In-situ Viscoelastic Gelation & Biological Plug Formation

- **Accelerated Coagulation Cascade:** Upon delivery of the granular 'Branzino Pepper' into the wound tract, immediate contact with whole blood triggers active catalysts (e.g., Thrombin or high-concentration Chitosan derivatives). This induces rapid platelet aggregation, facilitating a state of hyper-hemostasis that reduces physiological clotting time to a matter of seconds.
- **Fibrin Matrix Polymerization:** Beyond simple cessation of flow, the system orchestrates the polymerization of the effused blood into a dense, interconnected fibrin meshwork.
- **Viscoelastic Bio-plug Transition:** The intravascular and extravasated blood undergoes a definitive phase transition from a liquid state to a macroscopic viscoelastic gel (solidified hematoma). This high-volume thrombus functions as an overwhelming physical packing material, mechanically occluding ruptured arterial and venous vessels.

2. Mechanism of Lubrication: Blood-Activated Anti-Adhesion Lubricating Interface

- **Moisture-Triggered Release Coating:** Hydrophilic polymers (e.g., Polyethylene Glycol - PEG) integrated into the powder react with the aqueous component of the plasma to instantly synthesize a low-friction lubricating film.
- **Prevention of Tissue Integration:** This film establishes a non-adherent, anti-adhesion interface between the peripheral surface of the solidified blood mass and the internal wound walls (soft tissue). This biochemically inhibits the physical integration of the coagulated mass or subsequent fillers with myocytes and fibroblasts.
- **Shear Stress Dissipation:** The lubricating layer minimizes frictional force exerted on the surrounding tissue during external pressure application, thereby preventing secondary tissue laceration often caused by excessive volumetric expansion.

3. Synergistic Interaction: Hydrodynamic Synergy with Polyurethane Foam Expansion

- **Dual-Chamber Pressure Transfer:** Following the docking of the protective shield (e.g., knee protector), the injected polyurethane foam expands against the lubricated hemostatic mass rather than making direct contact with the denuded tissue.
- **Incompressible Fluid Dynamics:** Since the gelated blood mass possesses the properties of an incompressible fluid, it ensures a 100% uniform transmission of the pneumatic pressure generated by the expanding foam to the deep-seated vascular ruptures, achieving a total tamponade effect.
- **Defense Against Foam Embolism:** The solidified biological plug serves as a structural barrier, preventing the expanding foam from infiltrating the vascular lumen, thereby eliminating the risk of fatal foam-induced embolism.

4. Post-Trauma Management: Non-Invasive En Bloc Extraction & Irrigation

- **En Bloc Removal:** Upon arrival at a definitive care facility (ROLE 2/3), the administration of sterile saline by the surgical team maximizes the activity of the anti-adhesion interface. The combined mass of foam and clotted blood can be extracted via simple traction—similar to a cork—without the need for traumatic tissue scraping.
- **Minimization of OR Prep Time:** The absence of residual debris within the wound bed allows the surgical team to bypass extensive sharp debridement. This facilitates an immediate transition to definitive vascular repair, significantly increasing the probability of patient survival and limb salvage.

BRANZINO TACTICAL FIELD CUSTOMIZATION PROTOCOL

"A millimeter prepared in downtime saves ten minutes in combat."

PHASE 1: Anthropometric Measurement & Data Derivation

- **Objective:** To determine the precise relief slit (exhaust orifice) length tailored to the individual soldier's specific limb anatomy.
- **Circumference Measurement ($C = 2\pi r$):** During non-combat maintenance intervals, measure the exact circumference of the proximal thigh (inguinal region) and the brachium (upper arm) using a metric tape or high-tensile paracord.
- **D-Chart Calibration:** Input the recorded dimensions into the medical 'D-Chart' to calculate the 'Optimal Incision Length (mm)' for each anatomical site.

PHASE 2: Hybrid Pre-setting Configuration

A. Hard-Wired Critical Zones (Inguinal & Brachial Arteries)

- **Tactical Rationale:** High-lethality zones with a 60–120 second window for exsanguination.
- **Protocol:** These zones must be physically pre-scored (pre-cut) to the exact calculated D-Chart length. This ensures that the relief slit bursts open autonomously once the internal pressure reaches the threshold.

B. Soft-Wired Variable Zones (Axillary & Articular Junctions)

- **Tactical Rationale:** Regions where required tamponade pressure varies based on wound depth and angle.
- **Protocol:** Do not execute a pre-cut. Delineate the calculated limit line clearly using a permanent surgical marker. During trauma, execute the incision on-site along the marker.

PHASE 3: The "No-Slip" V-Groove Mechanical Guide

- **Objective:** To establish a physical guidance track that prevents blade deviation caused by tremors.
- **Protocol:** Carve a superficial 'V-groove' (micro-channel) directly over the permanent marker lines. It serves as a mechanical rail; even with severe tremors, the soldier can engage the combat knife into the channel, ensuring a linear trajectory.

PHASE 4: Combat Execution Sequence

1. **Trauma Identification:** Rapidly identify the injury site.
2. **Stabilization & Docking:** Secure the bite-tab to deploy the occlusive film, cover the wound, and lock clips.
3. **Incision (Conditional):** Bypass cutting phase for pre-scored zones. For marker zones, engage knife into V-groove.
4. **Hemostatic Injection:** Discharge total volume (100%) of the expanding foam.
5. **Egress:** Confirm venting of excess foam through the slit, then immediately proceed with extraction (Shoot & Run).

WARNING / DANGER

[Medical Contraindication]: NEVER execute an incision shorter than the designated 'Minimum Safety Line (Red Line)'. An insufficient exhaust diameter will result in catastrophic intracavitary over-pressurization, leading to secondary mechanical tissue necrosis and internal trauma.

Branzino System vs. Existing Technology Correspondence Table

Branzino Component	Matching Existing Technology	Remarks
1. Irrigation Spray	Pressurized Saline/Chlorhexidine	Used for mechanical wound debridement.
2. Coagulant Powder	Chitosan Granules (Celox / QuikClot)	Induces immediate gelation (Seonji-style).
3. Release Agent	PEG/Silicone Anti-adhesion Spray	Prevents post-surgical tissue adhesion.
4. Lubricated Gauze	XStat / Lubricated Packing Gauze	Syringe-driven for non-compressible hemorrhage.
5. Vinyl Adhesive Tape	Occlusive Dressing / Tactical Adhesives	High-tensile strength cutaneous adhesion.
6. Knee Protector	Injection Molded TPE / 3D Printed Nylon	High-durability tactical gear materials.
7. Hemostatic Foam	ResQFoam (DARPA) / PU Foam	Expanding foam for intracavitary tamponade.
8. Precision Scoring	Laser Scoring / Perforation Technology	Controlled material separation technology.

Technical Foundations and Deep Dive

1. **Primary Irrigation Spray:** Tactical systems like 'Irri-Gator' are established in field medicine for hydraulic debridement.
2. **Chemical Coagulation:** Products like 'Celox' (chitosan-based) promote rapid platelet aggregation independent of the body's natural cascade.
3. **Medical-Grade Release Agents:** Bio-resorbable barriers like 'Seprafilm' are utilized to prevent tissue adhesion.
4. **Tampon-style Syringe Applicators:** The 'XStat' device represents the state-of-the-art in injecting compressed cellulose sponges.
5. **Protective Shielding:** Gear from firms like Crye Precision employs high-performance Nylon/TPE composites.
6. **Single-use Expanding Hemostatic Foam:** The DARPA-supported 'ResQFoam' is the central equivalent, expanding 30x volume within 60s.

COMPLIANCE WARNING FOR MANUFACTURERS

"Any implementation must ensure 100% interoperability. Proprietary locking mechanisms or intentional incompatibility with DIY/3rd-party components are strictly prohibited under this license. The mission is survival, not monopoly."